

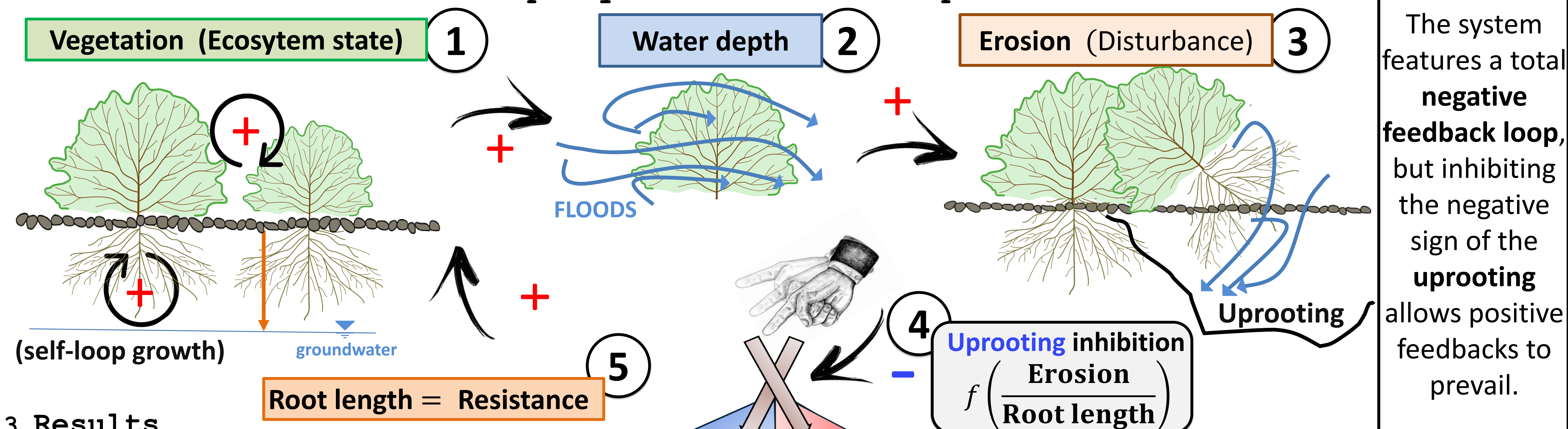
Feedback Interplay Governs River Ecosystem Chaos and Multi-Stability

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1. Introduction

Understanding how feedbacks between water, sediment, and vegetation interact to govern river dynamics remains limited. This knowledge is crucial for predicting river behavior and preserving climate-resilient ecosystems.

2. Numerical models and ecomorphodynamic feedback loop.



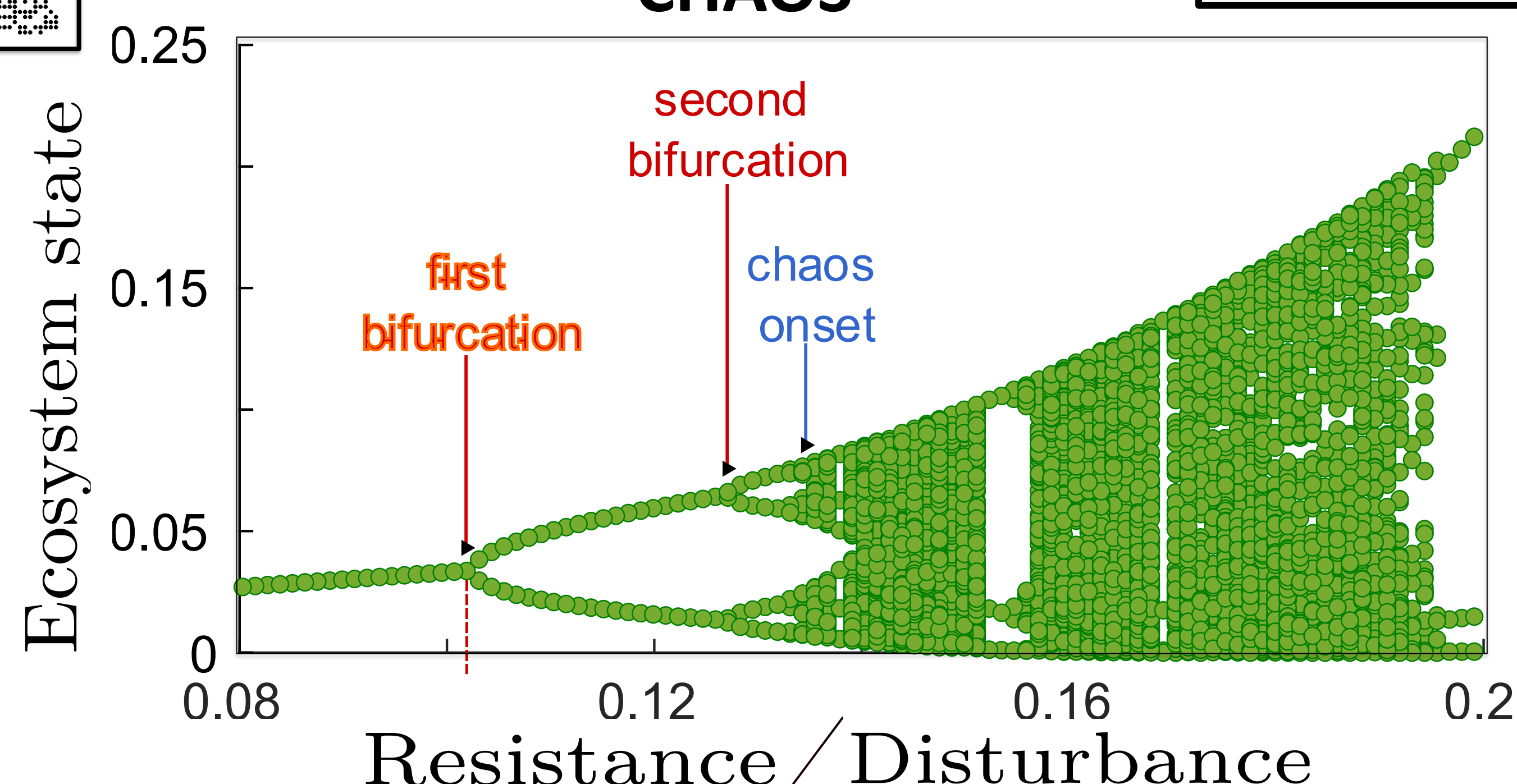
3. Results

NEGATIVE FEEDBACK LOOP

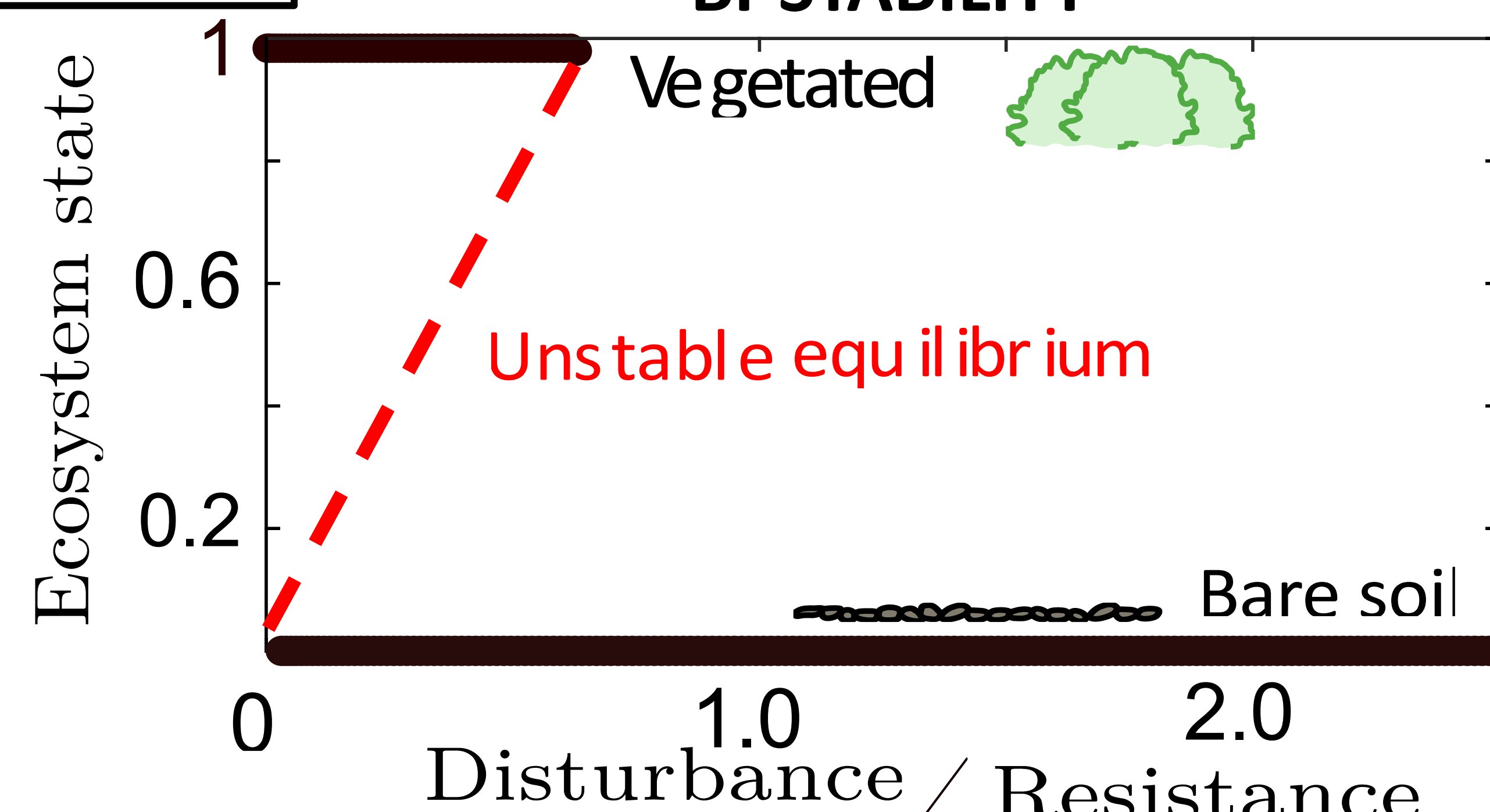
POSITIVE FEEDBACKS

Non-spatial model

CHAOS

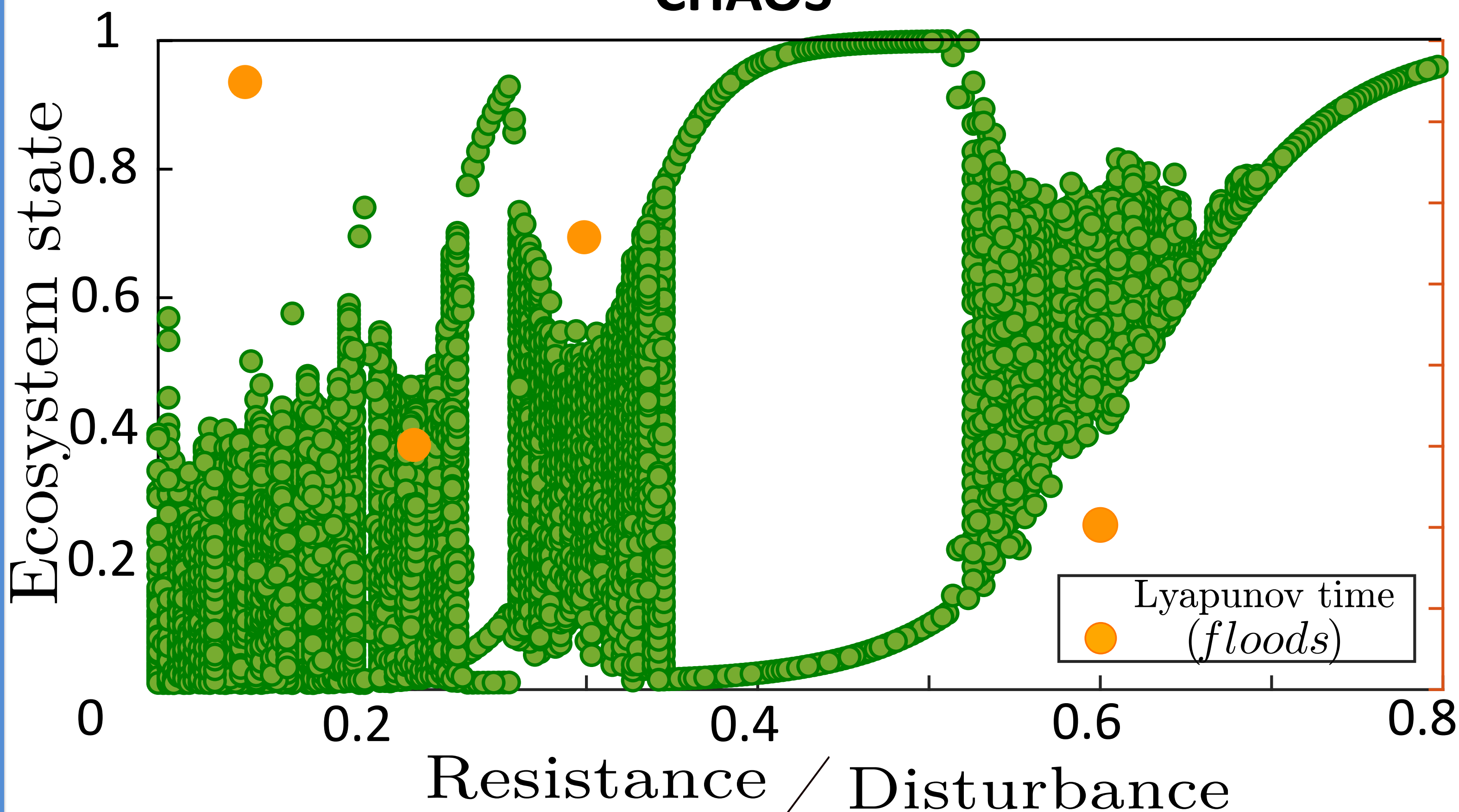


BI-STABILITY

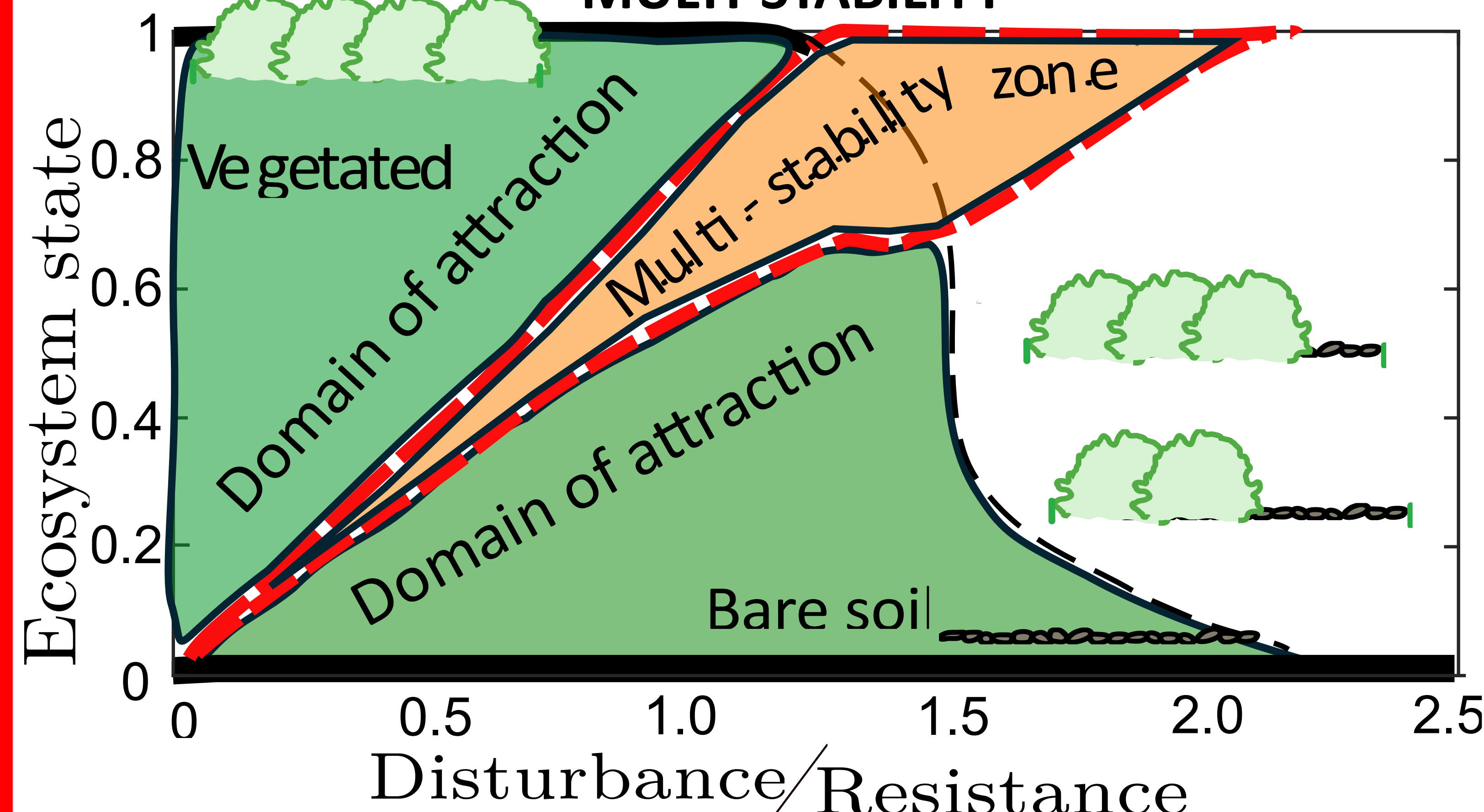


Spatial model

CHAOS



MULTI-STABILITY



4. Conclusions

- The **negative feedback loop** among vegetation, flow, and sediments generates oscillations and **chaotic** behavior.
- In chaotic regime the system is **not predictable** after **few floods**.

- With **positive feedback**, the system reaches steady state: non-spatial shows bi-stability, **spatial is multi-stable**.
- In the spatial model critical transitions and tipping points are smoothed and the **ecosystem is more resistant and resilient**.